

Product Description

GFTP-400 LP Low Dielectric Gap Pad is a thermally conductive interface material designed to provide efficient heat transfer between heat-generating components and heat sinks, while offering **low dielectric constant performance** for high-frequency and signal-sensitive electronic applications. The material is suitable for use in electronic assemblies where thermal management and electrical insulation are required, while conforming to surface irregularities to maintain consistent contact and minimizing mechanical stress on sensitive components. GFTP-400 LP is based on a silicone-filled composite engineered to provide reliable thermal conduction, low dielectric constant, and stable electrical insulation, while delivering good surface wettability, low compression force, and effective gap filling performance across a range of assembly conditions.

Key Features

- **Low dielectric constant**
- Conformable for effective gap filling
- Low compression force for reduced stress
- Stable performance over wide temperatures
- Self-adhesive for easy assembly

Applications

- RF and high-frequency components requiring low dielectric constant
- High-speed digital systems where signal integrity is critical
- Telecommunications and networking hardware
- Dense PCB assemblies with signal-sensitive layouts
- Automotive electronics with radar and communication links

Application Guidelines

- Avoid exposure to open flame and direct sunlight.
- Ensure application surfaces are clean.

Storage and Shelf Life

- Store at 8°C to 28°C in a dry environment (RH<70%).
- Shelf life is 12 months from the date of manufacture.

Packaging Information

- Available in thicknesses from 0.5 mm to 10.0 mm, with additional thicknesses available upon request
- Supplied in standard sheet formats and custom die-cut parts for specific application requirements
- Custom sizes and configurations supported to meet design and manufacturing needs

Property	Value	Test Method
Color	Purple	Visual
Hardness (Shore 00)	40	ASTM D2240
Thermal Conductivity (W/(m·K))	4.0	ASTM D5470
Thickness (mm)	0.5-10	--
Density (g/cm ³)	1.6	ASTM D792
Breakdown Voltage (kV/mm)	>10	ASTM D149
Operating Temperature Range (°C)	-50 to 200	--
Volume Resistivity (Ω·cm)	10 ¹⁴	ASTM D257
Flammability Rating	V-0	UL94
Dielectric Constant @ 5 GHz	3.8	ASTM D150

The data provided are typical values derived from laboratory testing and are intended for reference only. They should not be used as specifications or as a basis for design without independent validation.